2(a) Explain the various energy conversions that occur during each of the following events:

(i) A car moving on a rough horizontal road coming to a stop. [2]

(ii) A mass is attached to the bottom of a vertical spring. The mass is lowered slowly until it reaches an equilibrium position and remains stationary at that position. [3]

(b) The skateboarder in Fig. 2.1 below is 55 kg. She enters the ramp of height *h* at point A with a speed of 6.5 m s^{-1} and leaves the ramp vertically upwards at point B. The skateboarder experiences an average frictional force of 2.0 N by the ramp. The distance travelled between A and B is 2.0 m and *h* is 1.3 m.

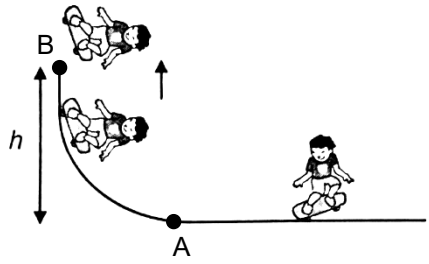

Fig. 2.1

(i) Determine the speed of the skateboarder as she exits point B. [3]

(ii) State and explain whether your answer to b(i) will be different if the ramp is replaced by the one shown in Fig. 2.2 below. The average frictional force by the ramp is still 2.0 N. The points A and B are in the same positions as in Fig. 2.1. [2]

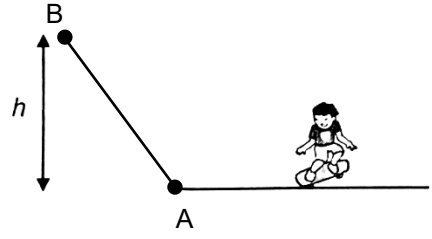

Fig. 2.2